

Expression Manipulation

cons-atom

Description: Constructs an expression (list) by adding an atom to the head of another expression.

Parameters:

- Head: The atom to be added to the beginning.
- Tail: The expression to add the atom on.

Return: A new expression.

Example:

```
!(cons-atom 1 (2 3)) ; Returns (1 2 3)
```

decons-atom

Description: Deconstructs an expression into its head and tail.

Parameters:

- Expression: The expression to be deconstructed.

Return: An expression containing the head and tail: `(Head Tail)`.

Example:

```
!(decons-atom (1 2 3)) ; Returns ((1) (2 3))
```

car-atom

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Description: Extracts the first atom of an expression.

Parameters:

- Expression: The expression to extract the first atom from.

Return: The first atom of the expression.

Example:

```
!(car-atom (1 2 3)) ; Returns 1
```

cdr-atom

Description: Extracts the tail of an expression (all atoms except the first).

Parameters:

- Expression: The expression to extract the tail from.

Return: The tail of the expression.

Example:

```
!(cdr-atom (1 2 3)) ; Returns (2 3)
```

size-atom

Description: Returns size of an expression.

Parameters:

- Expression: The expression whose size is to be determined..

Return: Size of an expression.

Example:

```
!(size-atom (1 2 3)) ; Returns 3
```

index-atom

Description: Returns atom from an expression using index or error if index is out of bounds

Parameters:

- Expression: The expression to extract from.
- Index: The index of the target atom.

Return: Atom from an expression in the position defined by index. Error if index is out of bounds

Example:

```
!(index-atom (1 2 3) 1) ; Returns 2
```